



United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Class Test-II Year: 2021 Trimester: Summer Program: BSCSE

Course: CSE 2215/CSI 217 Data Structure and Algorithms, Marks: 20

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There are Four questions. Answer all of them. Please submit assignment by 10/08/2021 within 11:55pm(night) to lms.

1. Show the mechanism of descending Order Heap Sort algorithm on the following data. [5]
Here DD means last two digits of your student ID.
10 8 20 DD 30
2. Discuss the time complexity of Selection Sort Algorithm [5]
3. Find the time complexity of the following algorithm [5]

```
int Fibonacci (int n){  
    if (n==1)  
        return 0;  
    else if (n==2)  
        return 1;  
    else  
        return Fibonacci(n-2) +Fibonacci (n-1);  
}
```
4. Prove that $T(n)=3n-2=\Theta(n)$ [5]